

# Implicit Differentiation and Inverse Functions

ANSWER KEY



## Implicit differentiation basics

- 1 Find  $\frac{dy}{dx}$  for  $x^2 + y^2 = 25$ .

$$2x + 2y \frac{dy}{dx} = 0, \text{ so } \frac{dy}{dx} = -\frac{x}{y}.$$

- 2 Find  $\frac{dy}{dx}$  for  $x^3 + y^3 = 6xy$ .

$$3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}; \frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2, \text{ so } \frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

- 3 Find  $\frac{dy}{dx}$  for  $\sin(xy) = x$ .

$$\cos(xy)(y + x \frac{dy}{dx}) = 1: y \cos(xy) + x \cos(xy) \frac{dy}{dx} = 1, \text{ so } \frac{dy}{dx} = \frac{1 - y \cos(xy)}{x \cos(xy)}.$$

---

## Tangent lines to implicit curves

- 4 Find the equation of the tangent line to  $x^2 + y^2 = 25$  at the point  $(3, 4)$ .

$$\frac{dy}{dx} = -\frac{x}{y}; \text{ at } (3, 4): \text{ slope} = -\frac{3}{4}. \text{ Tangent: } y - 4 = -\frac{3}{4}(x - 3), \text{ i.e. } y = -\frac{3}{4}x + \frac{25}{4}.$$

- 5 Find  $\frac{dy}{dx}$  at the point  $(1, 2)$  on the curve  $x^2y + y^3 = 10$ .

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0: \frac{dy}{dx}(x^2 + 3y^2) = -2xy, \text{ so } \frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}. \text{ At } (1, 2): \frac{-4}{1+12} = -\frac{4}{13}.$$

---

## Second derivatives via implicit differentiation

- 6 Find  $\frac{d^2y}{dx^2}$  for  $x^2 + y^2 = 1$ , expressing your answer in terms of  $x$  and  $y$ .

$$\frac{dy}{dx} = -\frac{x}{y}. \text{ Differentiate again: } \frac{d^2y}{dx^2} = -\frac{y - x \frac{dy}{dx}}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{1}{y^3} \text{ (using } x^2 + y^2 = 1).$$

---

## Derivatives of inverse functions

- 7 If  $f(x) = x^3 + 2x + 1$ , find  $(f^{-1})'(4)$ , using  $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$  and noting  $f(1) = 4$ .

$$f'(x) = 3x^2 + 2; f'(1) = 5. (f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{5}.$$

- 8 State the derivatives of the inverse trigonometric functions  $\sin^{-1} x$ ,  $\cos^{-1} x$ , and  $\tan^{-1} x$ .

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}; \frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}; \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}.$$

- 9 Differentiate  $f(x) = \tan^{-1}(3x)$ .

$$f'(x) = \frac{3}{1 + (3x)^2} = \frac{3}{1 + 9x^2}.$$

---

### Implicit differentiation with logarithmic and exponential terms

- 10 Find  $\frac{dy}{dx}$  for  $e^y = xy$ .

$$e^y \frac{dy}{dx} = y + x \frac{dy}{dx}; \frac{dy}{dx}(e^y - x) = y, \text{ so } \frac{dy}{dx} = \frac{y}{e^y - x}.$$

- 11 Find  $\frac{dy}{dx}$  for  $\ln(x + y) = x^2$ .

$$\frac{1 + \frac{dy}{dx}}{x + y} = 2x; 1 + \frac{dy}{dx} = 2x(x + y), \text{ so } \frac{dy}{dx} = 2x(x + y) - 1.$$

---

### Related applications of implicit differentiation

- 12 Find all points on the curve  $x^2 + xy + y^2 = 7$  where the tangent line is horizontal.

Differentiate:  $2x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$ ;  $\frac{dy}{dx} = \frac{-2x - y}{x + 2y}$ . Horizontal tangent needs  $-2x - y = 0$ , i.e.  $y = -2x$ .  
Substituting into the curve:  $x^2 + x(-2x) + (-2x)^2 = 7 \Rightarrow x^2 - 2x^2 + 4x^2 = 7 \Rightarrow 3x^2 = 7 \Rightarrow x = \pm\sqrt{7/3}$ .  
Points:  $(\pm\sqrt{7/3}, \mp 2\sqrt{7/3})$ .

---

### Finding vertical tangents implicitly

- 13 Find the points on the circle  $x^2 + y^2 = 9$  where the tangent line is vertical (i.e. where  $\frac{dx}{dy} = 0$ , or equivalently where  $\frac{dy}{dx}$  is undefined).

$\frac{dy}{dx} = -\frac{x}{y}$  is undefined when  $y = 0$ . Substituting into the circle:  $x^2 = 9$ , so  $x = \pm 3$ . Vertical tangents occur at  $(3, 0)$  and  $(-3, 0)$ .

---

### Inverse function derivatives from a graph or table

- 14 A function  $f$  is one-to-one and differentiable, with  $f(3) = 10$  and  $f'(3) = 4$ . Find  $(f^{-1})'(10)$ .

$$(f^{-1})'(10) = \frac{1}{f'(f^{-1}(10))} = \frac{1}{f'(3)} = \frac{1}{4} \text{ (since } f(3) = 10 \text{ means } f^{-1}(10) = 3).$$

- 15 If  $g(x) = x^5 + 2x - 3$ , find  $(g^{-1})'(-3)$ , first identifying the value  $a$  such that  $g(a) = -3$ .

$$g(0) = 0 + 0 - 3 = -3, \text{ so } a = 0. g'(x) = 5x^4 + 2; g'(0) = 2. (g^{-1})'(-3) = \frac{1}{g'(0)} = \frac{1}{2}.$$

---

### Implicit differentiation in an applied context

- 16 A conical tank has height and radius related by  $h = 2r$  at all times as it fills, so its volume is  $V = \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ . Using implicit differentiation with respect to time, find  $\frac{dV}{dt}$  in terms of  $r$  and  $\frac{dr}{dt}$ .
- Differentiating both sides with respect to  $t$ :  $\frac{dV}{dt} = \frac{2}{3}\pi(3r^2)\frac{dr}{dt} = 2\pi r^2\frac{dr}{dt}$ .

## A capstone implicit differentiation problem

- 17 Find  $\frac{dy}{dx}$  for the curve  $x^2y^2 + xy = 6$  at the point  $(1, 2)$ .

Differentiate:  $2xy^2 + x^2(2y)\frac{dy}{dx} + y + x\frac{dy}{dx} = 0$ :  $\frac{dy}{dx}(2x^2y + x) = -2xy^2 - y$ . At  $(1, 2)$ :  $\frac{dy}{dx}(4 + 1) = -4 - 2 = -6$ , so  $\frac{dy}{dx} = -\frac{6}{5}$ .

- 18 Find  $\frac{dy}{dx}$  for  $\cos(x - y) = y\sin x$ .

Differentiate:  $-\sin(x - y)(1 - \frac{dy}{dx}) = \frac{dy}{dx}\sin x + y\cos x$ :  $-\sin(x - y) + \sin(x - y)\frac{dy}{dx} = \sin x\frac{dy}{dx} + y\cos x$ .  
Collecting terms:  $\frac{dy}{dx}[\sin(x - y) - \sin x] = y\cos x + \sin(x - y)$ , so  $\frac{dy}{dx} = \frac{y\cos x + \sin(x - y)}{\sin(x - y) - \sin x}$ .

- 19 A satellite dish's cross-section satisfies  $4x^2 + y^2 = 16$  (an ellipse). Find  $\frac{dy}{dx}$  at the point  $(1, 2\sqrt{3})$  on the curve.

Differentiate:  $8x + 2y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = -\frac{4x}{y}$ . At  $(1, 2\sqrt{3})$ :  $\frac{dy}{dx} = -\frac{4}{2\sqrt{3}} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$ .

- 20 For the curve  $y^2 = x^3$  (a cuspidal cubic), complete each step:

a Find  $\frac{dy}{dx}$  implicitly.  $2y\frac{dy}{dx} = 3x^2$ , so  $\frac{dy}{dx} = \frac{3x^2}{2y}$ .

b Evaluate  $\frac{dy}{dx}$  at the point  $(4, 8)$ . At  $(4, 8)$ :  $\frac{dy}{dx} = \frac{3(16)}{2(8)} = \frac{48}{16} = 3$ .

c Find the equation of the tangent line at that point.  $y - 8 = 3(x - 4)$ , i.e.  $y = 3x - 4$ .

- 21 If  $f(x) = 2x^3 + x - 5$  is one-to-one and differentiable, find  $(f^{-1})'(-2)$ , first identifying the input  $a$  such that  $f(a) = -2$ .

$f(1) = 2 + 1 - 5 = -2$ , so  $a = 1$ .  $f'(x) = 6x^2 + 1$ ;  $f'(1) = 7$ .  $(f^{-1})'(-2) = \frac{1}{f'(1)} = \frac{1}{7}$ .

- 22 For the hyperbola  $xy = 6$ , complete each step:

a Find  $\frac{dy}{dx}$  using implicit differentiation.  $x\frac{dy}{dx} + y = 0$ , so  $\frac{dy}{dx} = -\frac{y}{x}$ .

b Confirm your answer matches what you'd get by first solving for  $y = \frac{6}{x}$  and differentiating directly. Direct:  $y = 6x^{-1}$ , so  $\frac{dy}{dx} = -6x^{-2} = -\frac{6}{x^2}$ . Substituting  $y = \frac{6}{x}$  into  $-\frac{y}{x}$ :  $-\frac{6/x}{x} = -\frac{6}{x^2}$ . Both match.

c Evaluate  $\frac{dy}{dx}$  at the point  $(2, 3)$ . At  $(2, 3)$ :  $\frac{dy}{dx} = -\frac{3}{2}$ .

## Multiple choice — easy 15

23 Find  $\frac{dy}{dx}$  for  $x^2 + 5y^2 = 126$  at the point  $(-1, 5)$ .

a.  $\frac{1}{25}$  ✓

b.  $\frac{1}{5}$

c. 1

d.  $-\frac{1}{25}$

$2x + 10y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{5y}$ . At  $(-1, 5)$ :  $\frac{1}{25} = \frac{1}{25}$ .

24 Find  $\frac{dy}{dx}$  for  $x^2 + 5y^2 = 30$  at the point  $(-5, 1)$ .

a. 5

b.  $-1$

c. 25

d. 1 ✓

$2x + 10y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{5y}$ . At  $(-5, 1)$ :  $\frac{5}{5} = 1$ .

25 Find  $\frac{dy}{dx}$  for  $x^2 + 4y^2 = 20$  at the point  $(2, 2)$ .

a.  $-1$

b.  $\frac{1}{4}$

c.  $-4$

d.  $-\frac{1}{4}$  ✓

$2x + 8y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{4y}$ . At  $(2, 2)$ :  $\frac{-2}{8} = -\frac{1}{4}$ .

26 Find  $\frac{dy}{dx}$  for  $x^2 + y^2 = 41$  at the point  $(4, 5)$ .

a.  $\frac{4}{5}$

b.  $\frac{1}{5}$

c.  $-\frac{3}{10}$

d.  $-\frac{4}{5}$  ✓

$2x + 2y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{y}$ . At  $(4, 5)$ :  $\frac{-4}{5} = -\frac{4}{5}$ .

27 Find  $\frac{dy}{dx}$  for  $x^2 + y^2 = 20$  at the point  $(2, 4)$ .

a.  $-\frac{1}{2}$  ✓

b.  $\frac{1}{2}$

c.  $-1$

d.  $0$

$2x + 2y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{y}$ . At  $(2, 4)$ :  $\frac{-2}{4} = -\frac{1}{2}$ .

28 Find  $\frac{dy}{dx}$  for  $x^2 + 5y^2 = 141$  at the point  $(4, -5)$ .

a.  $-\frac{4}{25}$

b.  $\frac{4}{5}$

c.  $\frac{4}{25}$  ✓

d.  $4$

$2x + 10y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{5y}$ . At  $(4, -5)$ :  $\frac{-4}{-25} = \frac{4}{25}$ .

29 Find  $\frac{dy}{dx}$  for  $x^2 + y^2 = 29$  at the point  $(-5, 2)$ .

a.  $3$

b.  $\frac{5}{2}$  ✓

c.  $-\frac{5}{2}$

d.  $2$

$2x + 2y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{y}$ . At  $(-5, 2)$ :  $\frac{5}{2} = \frac{5}{2}$ .

30 Find  $\frac{dy}{dx}$  for  $x^2 + y^2 = 41$  at the point  $(-5, 4)$ .

a.  $\frac{9}{4}$

b.  $\frac{7}{4}$

c.  $-\frac{5}{4}$

d.  $\frac{5}{4}$  ✓

$2x + 2y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{y}$ . At  $(-5, 4)$ :  $\frac{5}{4} = \frac{5}{4}$ .

31 Find  $\frac{dy}{dx}$  for  $x^2 + y^2 = 13$  at the point  $(-3, 2)$ .

a.  $-\frac{3}{2}$

b. 2

c.  $\frac{3}{2}$  ✓

d. 1

$2x + 2y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{y}$ . At  $(-3, 2)$ :  $\frac{3}{2} = \frac{3}{2}$ .

32 Find  $\frac{dy}{dx}$  for  $x^2 + 4y^2 = 40$  at the point  $(-2, -3)$ .

a.  $\frac{1}{6}$

b.  $-\frac{2}{3}$

c.  $-\frac{8}{3}$

d.  $-\frac{1}{6}$  ✓

$2x + 8y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{4y}$ . At  $(-2, -3)$ :  $\frac{-2}{-12} = -\frac{1}{6}$ .

33 Find  $\frac{dy}{dx}$  for  $x^2 + 4y^2 = 8$  at the point  $(2, 1)$ .

a.  $-\frac{1}{2}$  ✓

b. -2

c. -8

d.  $\frac{1}{2}$

$2x + 8y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{4y}$ . At  $(2, 1)$ :  $\frac{-2}{4} = -\frac{1}{2}$ .

34 Find  $\frac{dy}{dx}$  for  $x^2 + 3y^2 = 28$  at the point  $(1, 3)$ .

a.  $\frac{1}{9}$

b.  $-1$

c.  $-\frac{1}{9}$  ✓

d.  $-\frac{1}{3}$

$2x + 6y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{3y}$ . At  $(1, 3)$ :  $\frac{-1}{9} = -\frac{1}{9}$ .

35 Find  $\frac{dy}{dx}$  for  $x^2 + 3y^2 = 64$  at the point  $(4, -4)$ .

a.  $-\frac{1}{3}$

b.  $3$

c.  $\frac{1}{3}$  ✓

d.  $1$

$2x + 6y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{3y}$ . At  $(4, -4)$ :  $\frac{-4}{-12} = \frac{1}{3}$ .

36 Find  $\frac{dy}{dx}$  for  $x^2 + 3y^2 = 28$  at the point  $(5, 1)$ .

a.  $-5$

b.  $-15$

c.  $\frac{5}{3}$

d.  $-\frac{5}{3}$  ✓

$2x + 6y\frac{dy}{dx} = 0$ , so  $\frac{dy}{dx} = \frac{-x}{3y}$ . At  $(5, 1)$ :  $\frac{-5}{3} = -\frac{5}{3}$ .

37 Find  $\frac{dy}{dx}$  for  $x^2 + 4y^2 = 89$  at the point  $(-5, -4)$ .

- a.  $\frac{5}{16}$
- b.  $-5$
- c.  $-\frac{5}{4}$
- d.  $-\frac{5}{16}$  ✓

$$2x + 8y \frac{dy}{dx} = 0, \text{ so } \frac{dy}{dx} = \frac{-x}{4y}. \text{ At } (-5, -4): \frac{5}{-16} = -\frac{5}{16}.$$

---

### Multiple choice — medium 15

38 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 12$  at the point  $(2, -2)$ .

- a.  $-1$
- b.  $2$
- c.  $1$  ✓
- d.  $0$

$$y^2 + x(2y) \frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (2, -2): \frac{-8}{-8} = 1.$$

39 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 70$  at the point  $(5, -3)$ .

- a.  $\frac{1}{30}$
- b.  $\frac{19}{15}$
- c.  $\frac{19}{30}$  ✓
- d.  $-\frac{19}{30}$

$$y^2 + x(2y) \frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (5, -3): \frac{-19}{-30} = \frac{19}{30}.$$

40 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 26$  at the point  $(1, -5)$ .

- a.  $-\frac{27}{10}$
- b.  $-\frac{23}{10}$
- c.  $\frac{27}{10}$  ✓
- d.  $\frac{27}{5}$

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (1, -5): \frac{-27}{-10} = \frac{27}{10}.$$

41 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 12$  at the point  $(-4, 1)$ .

- a.  $\frac{7}{8}$
- b.  $-\frac{7}{4}$
- c.  $-\frac{9}{8}$
- d.  $-\frac{7}{8}$  ✓

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (-4, 1): \frac{7}{8} = -\frac{7}{8}.$$

42 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 17$  at the point  $(1, 4)$ .

- a.  $\frac{9}{4}$
- b.  $-\frac{9}{4}$  ✓
- c.  $\frac{7}{4}$
- d.  $-\frac{9}{2}$

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (1, 4): \frac{-18}{8} = -\frac{9}{4}.$$

43 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 5$  at the point  $(-5, 2)$ .

- a.  $-\frac{3}{5}$
- b.  $-\frac{3}{10}$  ✓
- c.  $-\frac{7}{10}$
- d.  $\frac{3}{10}$

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (-5, 2): \frac{6}{-20} = -\frac{3}{10}.$$

44 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = -39$  at the point  $(-3, 4)$ .

a.  $\frac{5}{6}$

b.  $-\frac{11}{12}$

c.  $\frac{5}{12}$  ✓

d.  $-\frac{5}{12}$

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (-3, 4): \frac{-10}{-24} = \frac{5}{12}.$$

45 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 12$  at the point  $(-4, -1)$ .

a.  $\frac{7}{4}$

b.  $\frac{7}{8}$  ✓

c.  $\frac{9}{8}$

d.  $-\frac{7}{8}$

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (-4, -1): \frac{7}{8} = \frac{7}{8}.$$

46 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 21$  at the point  $(3, 2)$ .

a.  $-\frac{1}{6}$

b.  $-\frac{5}{3}$

c.  $\frac{5}{6}$

d.  $-\frac{5}{6}$  ✓

$$y^2 + x(2y)\frac{dy}{dx} + 2x = 0: \frac{dy}{dx}(2xy) = -y^2 - 2x, \text{ so } \frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}. \text{ At } (3, 2): \frac{-10}{12} = -\frac{5}{6}.$$

47 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 45$  at the point  $(5, 2)$ .

a.  $-\frac{7}{10}$  ✓

b.  $\frac{7}{10}$

c.  $-\frac{7}{5}$

d.  $-\frac{3}{10}$

$y^2 + x(2y)\frac{dy}{dx} + 2x = 0$ :  $\frac{dy}{dx}(2xy) = -y^2 - 2x$ , so  $\frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}$ . At  $(5, 2)$ :  $\frac{-14}{20} = -\frac{7}{10}$ .

48 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = -55$  at the point  $(-5, -4)$ .

a.  $-\frac{3}{10}$

b.  $\frac{3}{20}$

c.  $\frac{13}{20}$

d.  $-\frac{3}{20}$  ✓

$y^2 + x(2y)\frac{dy}{dx} + 2x = 0$ :  $\frac{dy}{dx}(2xy) = -y^2 - 2x$ , so  $\frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}$ . At  $(-5, -4)$ :  $\frac{-6}{40} = -\frac{3}{20}$ .

49 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 54$  at the point  $(2, 5)$ .

a.  $\frac{29}{20}$

b.  $\frac{21}{20}$

c.  $-\frac{29}{10}$

d.  $-\frac{29}{20}$  ✓

$y^2 + x(2y)\frac{dy}{dx} + 2x = 0$ :  $\frac{dy}{dx}(2xy) = -y^2 - 2x$ , so  $\frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}$ . At  $(2, 5)$ :  $\frac{-29}{20} = -\frac{29}{20}$ .

50 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 6$  at the point  $(-3, -1)$ .

a.  $\frac{7}{6}$

b.  $-\frac{5}{6}$

c.  $\frac{5}{6}$  ✓

d.  $\frac{5}{3}$

$y^2 + x(2y)\frac{dy}{dx} + 2x = 0$ :  $\frac{dy}{dx}(2xy) = -y^2 - 2x$ , so  $\frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}$ . At  $(-3, -1)$ :  $\frac{5}{6} = \frac{5}{6}$ .

51 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 54$  at the point  $(2, -5)$ .

a.  $\frac{29}{20}$  ✓

b.  $-\frac{21}{20}$

c.  $\frac{29}{10}$

d.  $-\frac{29}{20}$

$y^2 + x(2y)\frac{dy}{dx} + 2x = 0$ :  $\frac{dy}{dx}(2xy) = -y^2 - 2x$ , so  $\frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}$ . At  $(2, -5)$ :  $\frac{-29}{-20} = \frac{29}{20}$ .

52 Find  $\frac{dy}{dx}$  for  $xy^2 + x^2 = 0$  at the point  $(-4, 2)$ .

a.  $-\frac{1}{2}$

b.  $-\frac{3}{4}$

c.  $\frac{1}{4}$

d.  $-\frac{1}{4}$  ✓

$y^2 + x(2y)\frac{dy}{dx} + 2x = 0$ :  $\frac{dy}{dx}(2xy) = -y^2 - 2x$ , so  $\frac{dy}{dx} = \frac{-y^2 - 2x}{2xy}$ . At  $(-4, 2)$ :  $\frac{4}{-16} = -\frac{1}{4}$ .

---

### Multiple choice — hard 15

53 If  $f(x) = 2x^3 + 6x + 5$  is one-to-one and differentiable, find  $(f^{-1})'(5)$ , using  $f(0) = 5$ .

a.  $\frac{1}{6}$  ✓

b.  $\frac{1}{7}$

c.  $-\frac{1}{6}$

d. 6

$f'(x) = 6x^2 + 6$ ;  $f'(0) = 6(0)^2 + 6 = 6$ .  $(f^{-1})'(5) = \frac{1}{f'(0)} = \frac{1}{6} = \frac{1}{6}$ .

54 If  $f(x) = 3x^3 - 4x + 3$  is one-to-one and differentiable, find  $(f^{-1})'(-66)$ , using  $f(-3) = -66$ .

a.  $-\frac{1}{77}$

b. 77

c.  $\frac{1}{77}$  ✓

d.  $\frac{1}{78}$

$$f'(x) = 9x^2 - 4; f'(-3) = 9(-3)^2 - 4 = 77. (f^{-1})'(-66) = \frac{1}{f'(-3)} = \frac{1}{77} = \frac{1}{77}.$$

55 If  $f(x) = 2x^3 + 6x - 3$  is one-to-one and differentiable, find  $(f^{-1})'(69)$ , using  $f(3) = 69$ .

a. 60

b.  $-\frac{1}{60}$

c.  $\frac{1}{61}$

d.  $\frac{1}{60}$  ✓

$$f'(x) = 6x^2 + 6; f'(3) = 6(3)^2 + 6 = 60. (f^{-1})'(69) = \frac{1}{f'(3)} = \frac{1}{60} = \frac{1}{60}.$$

56 If  $f(x) = x^3 + x + 3$  is one-to-one and differentiable, find  $(f^{-1})'(13)$ , using  $f(2) = 13$ .

a. 13

b.  $-\frac{1}{13}$

c.  $\frac{1}{13}$  ✓

d.  $\frac{1}{14}$

$$f'(x) = 3x^2 + 1; f'(2) = 3(2)^2 + 1 = 13. (f^{-1})'(13) = \frac{1}{f'(2)} = \frac{1}{13} = \frac{1}{13}.$$

57 If  $f(x) = 3x^3 - 4x + 2$  is one-to-one and differentiable, find  $(f^{-1})'(2)$ , using  $f(0) = 2$ .

a.  $-\frac{1}{4}$  ✓

b.  $\frac{1}{4}$

c.  $-\frac{1}{3}$

d. -4

$$f'(x) = 9x^2 - 4; f'(0) = 9(0)^2 - 4 = -4. (f^{-1})'(2) = \frac{1}{f'(0)} = \frac{1}{-4} = -\frac{1}{4}.$$

58 Find  $\frac{dy}{dx}$  for  $x^2y + y^3 = -68$  at the point  $(1, -4)$ .

a.  $-\frac{33}{98}$

b.  $-\frac{8}{49}$

c.  $\frac{8}{49}$  ✓

d.  $-\frac{8}{47}$

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0: \frac{dy}{dx}(x^2 + 3y^2) = -2xy, \text{ so } \frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}. \text{ At } (1, -4): \frac{8}{49} = \frac{8}{49}.$$

59 Find  $\frac{dy}{dx}$  for  $x^2y + y^3 = -10$  at the point  $(-3, -1)$ .

a. 0

b.  $-\frac{1}{2}$  ✓

c. -1

d.  $\frac{1}{2}$

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0: \frac{dy}{dx}(x^2 + 3y^2) = -2xy, \text{ so } \frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}. \text{ At } (-3, -1): \frac{-6}{12} = -\frac{1}{2}.$$

60 Find  $\frac{dy}{dx}$  for  $x^2y + y^3 = -16$  at the point  $(-2, -2)$ .

a.  $\frac{1}{2}$

b. -1

c. 1

d.  $-\frac{1}{2}$  ✓

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0: \frac{dy}{dx}(x^2 + 3y^2) = -2xy, \text{ so } \frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}. \text{ At } (-2, -2): \frac{-8}{16} = -\frac{1}{2}.$$

61 Find  $\frac{dy}{dx}$  for  $x^2y + y^3 = 17$  at the point  $(-4, 1)$ .

a.  $\frac{8}{19}$  ✓

b.  $-\frac{8}{19}$

c.  $\frac{8}{13}$

d.  $\frac{27}{19}$

$2xy + x^2\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$ :  $\frac{dy}{dx}(x^2 + 3y^2) = -2xy$ , so  $\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$ . At  $(-4, 1)$ :  $\frac{8}{19} = \frac{8}{19}$ .

62 Find  $\frac{dy}{dx}$  for  $x^2y + y^3 = 10$  at the point  $(-1, 2)$ .

a.  $-\frac{4}{13}$

b.  $-\frac{4}{11}$

c.  $\frac{4}{13}$  ✓

d.  $-\frac{5}{26}$

$2xy + x^2\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$ :  $\frac{dy}{dx}(x^2 + 3y^2) = -2xy$ , so  $\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$ . At  $(-1, 2)$ :  $\frac{4}{13} = \frac{4}{13}$ .

63 Find  $\frac{dy}{dx}$  for  $x^2 + xy + y^2 = 19$  at the point  $(-2, -3)$ .

a.  $\frac{7}{8}$

b.  $-\frac{7}{5}$

c.  $-\frac{7}{8}$  ✓

d.  $\frac{1}{8}$

$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$ :  $\frac{dy}{dx}(x + 2y) = -2x - y$ , so  $\frac{dy}{dx} = \frac{-2x - y}{x + 2y}$ . At  $(-2, -3)$ :  $\frac{7}{-8} = -\frac{7}{8}$ .

64 Find  $\frac{dy}{dx}$  for  $x^2 + xy + y^2 = 1$  at the point  $(1, -1)$ .

a.  $-3$

b.  $-1$

c.  $1$  ✓

d.  $\frac{1}{2}$

$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$ :  $\frac{dy}{dx}(x + 2y) = -2x - y$ , so  $\frac{dy}{dx} = \frac{-2x - y}{x + 2y}$ . At  $(1, -1)$ :  $\frac{-1}{-1} = 1$ .

65 Find  $\frac{dy}{dx}$  for  $x^2 + xy + y^2 = 7$  at the point  $(3, -2)$ .

a.  $-8$

b.  $\frac{9}{2}$

c.  $4$  ✓

d.  $-4$

$$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0: \frac{dy}{dx}(x + 2y) = -2x - y, \text{ so } \frac{dy}{dx} = \frac{-2x - y}{x + 2y}. \text{ At } (3, -2): \frac{-4}{-1} = 4.$$

66 Find  $\frac{dy}{dx}$  for  $x^2 + xy + y^2 = 7$  at the point  $(-1, -2)$ .

a.  $0$

b.  $-\frac{4}{5}$  ✓

c.  $-\frac{4}{3}$

d.  $\frac{4}{5}$

$$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0: \frac{dy}{dx}(x + 2y) = -2x - y, \text{ so } \frac{dy}{dx} = \frac{-2x - y}{x + 2y}. \text{ At } (-1, -2): \frac{4}{-5} = -\frac{4}{5}.$$

67 Find  $\frac{dy}{dx}$  for  $x^2 + xy + y^2 = 7$  at the point  $(1, 2)$ .

a.  $-\frac{4}{3}$

b.  $-\frac{4}{5}$  ✓

c.  $0$

d.  $\frac{4}{5}$

$$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0: \frac{dy}{dx}(x + 2y) = -2x - y, \text{ so } \frac{dy}{dx} = \frac{-2x - y}{x + 2y}. \text{ At } (1, 2): \frac{-4}{5} = -\frac{4}{5}.$$